Zihe (Zach) Liu

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Distributed Reinforcement Learning (RL) for LLMs, Long-horizon Alignment

EDUCATION

University of Cambridge

Master of Engineering, Information and Computer Engineering — *July 2026 - Present*

Masters Thesis: *Second Order Forward Optimization for Natural Policy Gradients*

- Develop efficient Gauss-Newton optimizer for RL, on custom OCaml ML compiler and AD stack (JAX-like)

Bachelor of Arts, Information and Computer Engineering — First Class Honors — *Sep 2023 - June 2026*

Coursework: RL, Deep Learning, Probabilistic Inference, Information Theory, Optimization, Computer Architecture

- David Newland Prize for Design (best undergrad project)

EXPERIENCE

ERA: AI — **Cambridge, UK**

Research Fellow — Mentor: Elizaveta Tennant, *Google DeepMind* — *July 2026 - Present*

- AI research program (top 0.5% of 6,000+ applicants) working on long-horizon RL for LLM alignment
- Develop multi-turn RL algorithm to reduce ethical violations 17% at 2× training horizon (80K tokens, 128 turns)
- Scale distributed RL in verl (FSDP, async vLLM) on *4xB300*. Fused GatedDeltaNet kernels raised MFU 1.75%→27%
- Traced GPU hang to FlashInfer kernel race condition, where fixing it enabled 4× higher inference throughput

Amazon — **London, UK**

Software Development Engineer Intern — *July - Oct 2025*

- Built ML *(CatBoost)* triaging system to root-cause fatal Prime Video errors, with *AWS Bedrock* LLM log analysis
- Scaled throughput 100× (3→300 sessions/cluster, 20GB logs) by batching SQL queries on *AWS Lambda, Athena*
- Refactored monolithic *AWS CDK* stack, and unblocked a *Fargate* pipeline stalling deployment for 4 months
- *Turned down full-time return offer to work on RL research*

UCL Centre for AI — **London, UK**

Student Researcher — Mentor: Peter Hayes, *Humanloop (now Anthropic)* — *Feb - May 2025*

- Developed *Critique Direct Preference Optimization*: generative critique for sample-efficient DPO | Blog
- Distributed training on *Qwen-2.5* 3B models with *16x H100 GPUs*, on UCL HPC facility | GitHub

Singapore AI Safety Institute, Infocomm Media Development Authority — **Singapore**

Research Engineer Intern — *Dec 2024 - July 2025*

- Research industry evaluation metrics for LLM reliability and robustness | Report
- Benchmark hallucination detectors on *HaluBench* (simple CoT LLM-judge outperforms all) | GitHub

Alignment Research Engineer Accelerator (ARENA) — **London, UK**

Research Engineer — 1 of 2 undergrads in a mostly-PhD cohort of ∼20 — *Sep - Oct 2024*

- Built GPT-2 LLM from scratch, reverse-engineered its internal reasoning circuits, and fine-tuned via RLHF (PPO)

Cambridge AI Safety Hub — **Cambridge, UK**

Operations Lead, Student Researcher — *Oct 2023 - Present*

- Organized AI safety workshops, hackathons, research reading groups on interpretability, alignment | R1 Talk

Changi Airport Group — **Singapore**

Machine Learning Engineer Intern — *July - Aug 2023*

- Trained P2PNet (object detection CNN model) to localize and count luggage trolleys *(MAE decreased by 5x compared to previous systems)*. Quantized model with ONNX to deploy on edge (Nvidia Jetson Nano)

Republic of Singapore Air Force — **Singapore**

Security Trooper, Corporal First Class — *Jan 2021 - Jan 2023*

- Led Immediate Response Force team (2IC) to win APGC Eagles Challenge 2022. Best Airman Award.

PUBLICATIONS

Z. Liu, D. Le, E. Tennant. Multi-Turn RL for Long-Horizon Moral Alignment. *NeurIPS 2026 AI4GOOD Workshop*

J. Robinson, L. Franceschetti, M. Lovett, D. Le, K. Sen, N. Narayanan, **Z. Liu**, E. Tennant. Reasoning–Behavior Coherence Beyond the Rewarded Context. *NeurIPS 2026 AIWILD*

M. Saghir, N. Raghavendra, **Z. Liu**, E. Gunter. NGD converges to less degenerate solutions than SGD. *arXiv*

AWARDS

SG Digital AI Scholarship | 500k$+ Full undergraduate funding, for exceptional Singaporeans pursuing AI overseas

GB3 RISC-V CPU Competition 2026 | *1st of 8 teams.* Optimize tiny CPU with instruction cache, lookaheads | GitHub

SF2 Image Compression 2026 | *1st of 13 teams.* Built JPEG compression scheme with DCT, intra prediction | Results

Changi Airport Group Scholarship | Offered full undergraduate funding for fast-track graduate programme

SKILLS

Programming Languages: Python, C/C++, OCaml, Verilog/SystemVerilog, TypeScript/JavaScript, MATLAB, Java

ML Systems: verl, TRL, vLLM, PyTorch FSDP, DeepSpeed-ZeRO, NumPy, JAX

AWS/Cloud: CloudFormation, CDK, Fargate, Lambda, S3, Athena

Non-ML: HPC (Sun Grid Engine), FPGA/Arduino/Embedded Systems, Git, SQL